Open System Interconnection- 7layers

1. Physical layer is the first layer of the OSI model which understands binary (0 & 1). Ex- RJ45

2. Data link layer is the second layer of the OSI which understands frames, MAC addresses and ARP table
   combination of IP and MAC addresses is ARP

3. Network layer is the third layer of OSI which understands IP and perform routing. Ex- router

4. Transport layer is the fourth layer of the OSI layer which decides how my data will be transported. Ex- TCP / UDP

5. Session layer is the fifth layer of the OSI which decides how long session is established between the connection.
   (By default, session time out of TCP is 1 hour or 30 min , for UDP is 5 minutes/30sec/60sec/180 sec.)

6. Presentation layer is the sixth layer of the OSI which decides how my data is presented.
   Ex-jpg, html, mp4

7. Application layer is the last layer of the OSI which presents the data into readable format.

2:cwhNIie2RJqDGjSrcfs-fOU2HdxVLUAwuBSzMyWxdWAQLN4MiSyZ2SwyRn1jF_fAKqP8l0vEiYbX7D8XgNXkVg

Layer	Name	Primary Function	Examples
7	**Application**	User interface and network services.	HTTP, FTP, SMTP, DNS
6	**Presentation**	Data formatting, encryption, and compression.	SSL/TLS, JPEG, ASCII
5	**Session**	Manages communication sessions between hosts.	NetBIOS, RPC, Sockets
4	**Transport**	End-to-end communication, error recovery, flow control.	TCP, UDP
3	**Network**	Routing, logical addressing (IP), path determination.	IP, ICMP, Routers
2	**Data Link**	Physical addressing (MAC), error detection for hardware.	Ethernet, Wi-Fi (802.11), Switches
1	**Physical**	Raw bit stream transmission via physical medium.	Cables, Hubs, Fiber optics

ISP (core)-> router (distribution layer)-> switch (accessing layer)-> devices.

Switch

Switch is a layer 2 device which is used to connect the endpoint devices such as laptops, printers etc.

In a network, a **switch** is a smart hardware device that connects multiple devices (computers, printers, servers) within a **Local Area Network (LAN)**. It acts as a central hub where all these devices plug in so they can "talk" to each other.

Hub: Broadcasts to everyone (Inefficient and insecure).

Switch: Sends directly to the target (Efficient, fast, and secure).

In the switch we have 1 broadcast domain and Multiple collision domain.

L2 switch is switch, L3 used for both switch and routing (core)

VLAN is used to divide 1 broadcast domain into multiple small broadcast domain.

It is a logical grouping of devices within in the same network that allows them to communicate as if they are on a same local network , even if they are physically connected to different switches.

In switch by default VLAN1. All interfaces are part VLAN1

Inter VLAN is used to communicate between multiple VLAN

A **Trunk Port** is a single physical port on a switch or firewall configured to carry traffic for **multiple VLANs** simultaneously.

If a standard "Access Port" is a single-lane road for one specific neighborhood (VLAN), a "Trunk Port" is a multi-lane highway connecting two cities.

STP- layer 2 protocol which is used to avoid layer two loop. Default priority – 32768

low-0

Lowest mac will be the route bridge in STP priority of STP (low mac+priority)

Bridge datagram

STP (Spanning Tree Protocol) is a Layer 2 network protocol designed to prevent **switching loops**.

In a network with redundant links (multiple paths between switches), frames can circulate indefinitely, causing a "broadcast storm" that crashes the network. STP identifies these loops and strategically shuts down specific ports to create a single, loop-free path.

What exactly is blocked?

STP does not block the entire physical cable; it places specific **switch ports** into a **Blocking State**.

- **What is blocked:** Only **User Data** (standard traffic, broadcasts like ARP, and multicasts) is blocked.
- **What is allowed:** The port remains "active" only to listen for **BPDUs** (Bridge Protocol Data Units). These are STP's own management messages. If the main path fails, the blocked port hears the failure via BPDUs and transitions to "Forwarding" to restore connectivity.

Example: 4-Switch Redundant Topology

Imagine four switches connected in a square (a physical loop):

- **Switch A** is connected to B and D.
- **Switch B** is connected to A and C.
- **Switch C** is connected to B and D.
- **Switch D** is connected to A and C.

1. Election of the Root Bridge

The switches exchange BPDUs. The switch with the **lowest Bridge ID** (Priority + MAC address) becomes the **Root Bridge**. Let's say **Switch A** wins.

2. Determining Path Costs

Every other switch finds the "cheapest" (fastest) path to Switch A.

- **Switch B** uses its direct link to A (**Root Port**).
- **Switch D** uses its direct link to A (**Root Port**).

3. Identifying the Loop

Switch C has two ways to get to the Root (via B or via D). If all links are the same speed, Switch C will pick one as its **Root Port** (let's say the link to B).

4. The Block

To break the loop, STP must shut down the final link. The link between **Switch C and Switch D** is the redundant path.

- One side of the link (on Switch D) will stay as a **Designated Port** (Forwarding).

- The other side (on Switch C) will become the **Alternate Port** and enter the **Blocking State**.

Result: Traffic from C to A must now go through B. If the link A-B fails, STP will automatically unblock the C-D link to provide a new path.

BPDU consists of who it thinks root bridge ID, sender bridge ID, path cost, timer

Router

Routing is the process of moving data packets from a source network to a destination network across an internetwork. While switching happens inside a local network (using MAC addresses), routing happens between networks (using IP addresses).

A router acts like a traffic controller at a major highway intersection, looking at a packet's destination IP and consulting its Routing Table to decide which "road" (path) to send the packet down.
protocols – static , default , dynamic

Static –

has a single path only no other alternative need to mention manually

The network administrator manually enters every route into the router's table.

- **How it works:** You tell the router, "To get to network X, go through this specific exit port."
- **Pros:** Very secure, consumes zero bandwidth/CPU for calculation.
- **Cons:** Not scalable. If a cable breaks, the router doesn't know; it will keep sending data into a "dead end" until you manually change it.

Default

like a gateway everyone can go like 0.0.0.0/0

A special type of static route, often called the "Gateway of Last Resort."

- How it works: You tell the router, "If you don't know where the destination is, just send it out this interface."
- Usage: Used in home routers (e.g., your router sends everything it doesn't recognize out to your ISP).

Dynamic

have multiple path to reach destination, if a path fails, they calculate a new one on the fly. always follows shortest path.

Routers "talk" to each other to learn about the network topology automatically. If a path fails, they calculate a new one on the fly.

- How it works: Routers exchange messages (advertisements) about the networks they are connected to.
- Pros: Extremely scalable; handles network changes automatically.
- Cons: Consumes router CPU and bandwidth.

Protocol	**Type**	**Key Feature**
RIP	Distance Vector	Simple; uses "hop count" (max 15). Slow to update.
OSPF	Link-State	Fast, efficient; uses "cost" (bandwidth). Great for enterprise.
EIGRP	Hybrid	Cisco proprietary (originally); fast convergence and very smart.
BGP	Path Vector	The "Internet protocol." Connects huge organizations/ISPs.

Bgp uses port no 179 not a protocol number.

The Details

- RIP (Routing Information Protocol): The old-school protocol. It counts how many routers (hops) a packet must pass through. It is rarely used today because it is slow to recover from failures and capped at 15 hops.

- OSPF (Open Shortest Path First): Currently the standard for internal corporate networks. Every router maps the entire network "topology" and calculates the fastest path based on bandwidth. If a link goes down, it updates everyone almost instantly.
- EIGRP (Enhanced Interior Gateway Routing Protocol): A high-performance protocol developed by Cisco. It is often faster than OSPF at finding new routes and uses complex math to choose the "best" path.
- BGP (Border Gateway Protocol): This is what keeps the internet alive. It is an Exterior Gateway Protocol (EGP). It doesn't care about internal speeds; it cares about connecting "Autonomous Systems" (large blocks of the internet owned by ISPs or giants like Google). It is extremely stable but complex to configure.

---

3. How Routing Decisions are Made (Administrative Distance)

If a router learns about the *same* destination from two different protocols (e.g., one route from OSPF and one from RIP), it uses Administrative Distance (AD) to decide which one to trust. The lower the number, the more "trustworthy" the source.

- Directly Connected: 0 (Most trusted)
- Static Route: 1
- EIGRP: 90
- OSPF: 110
- RIP: 120

RIP (Routing Information Protocol) is one of the oldest and simplest distance-vector routing protocols. It works by having routers share their "knowledge" of the network with their neighbours at regular intervals to determine the shortest path to a destination.

AD-120

The Core Metric: Hop Count

In RIP, the "best" path is defined strictly by the number of routers (hops) a packet must pass through to reach its destination.

- Directly connected: 0 hops.
- Through one router: 1 hop.
- The Limit: RIP has a maximum hop count of 15.

- Unreachable: A hop count of 16 is considered "infinity" or **unreachable. This prevents packets from looping forever in small networks.**

How Routers "Talk" (The Mechanism)

1. Periodic Updates: Every 30 seconds, a RIP-enabled router broadcasts (RIPv1) or multicasts (RIPv2) its entire routing table to all its immediate neighbors.
2. Learning by Addition: When Router A receives an update from Router B, it says: *"If Router B can reach Network X in 2 hops, and I can reach Router B in 1 hop, then I can reach Network X in 3 hops."*
3. The Best Path Wins: If Router A later receives a different update saying it can reach Network X in only 2 hops through a different neighbor, it will update its table with the shorter path.

Ping is used to test the connectivity between source and destination. it works on echo request and echo reply. it uses ICMP.

Trace route is used to check how many hops my destination is far from my source.

Supports up to 30 hops only.traceroute works on by decrementing the TTL value

nslookup

DHCP works based on DORA

D-discover , O- offer, R- request , A- acknowledgement

NTP- Network Time Protocol, helps to synchronize all our network devices with same time zone

Why it's critical for Palo Alto:

- Log Correlation: Timestamps must match across the network for forensic analysis.
- Certificate Validation: VPNs (GlobalProtect) and SSL Decryption fail if the clock skew makes certificates appear expired or not yet valid.

- Update Handshakes: Connection to Palo Alto update servers for App-ID and Threat signatures requires synchronized time for SSL/TLS.

Configuration Example:

WebUI: Device > Setup > Services > NTP

- Address: pool.ntp.org or time.google.com

CLI:

Bash

set deviceconfig system ntp-servers primary-ntp-server ntp-server-address time.google.com

commit

Verification: > show ntp (Look for status: synced)

VTP – VLAN Trunking Protocol layer 2 is used to one switch to another

Synchronise VLAN switch.

3 modes – server (create, delete, modify VLAN info), client(cannot do anything , only receives from server), transparent (create, delete, modify VLAN but can't part with other switch also won't take updates from server and client mode)

Trunk port is used to transport VLAN info from 1 switch to another switch

- **Uplink (User to Network): Used for sending, uploading, and posting data. Characterized by user-generated traffic (e.g., uploading a video, sending an email).**
- **Downlink (Network to User): Used for receiving, streaming, and browsing. Generally higher volume/bandwidth than uplink, such as watching Netflix, browsing the web, or downloading files.**

Firewall can be hardware or software which will block all the traffic unless or until we apply rule to allow it. Rules/access list/policies

Access control list is used to filter traffic 2 types of ACL (standard & extended AL)

Standard ACLs filter traffic based only on the source IP address (range 1–99, 1300–1999) and are placed close to the destination. Extended ACLs (range 100–199, 2000–

2699) offer granular control by filtering based on source/destination IPs, protocols (IP, TCP, UDP), and port numbers, and are placed close to the source

Feature	Standard ACL	Extended ACL
Check criteria	Source IP only	Source IP, Destination IP, Protocol, Port
Precision	Low	High
Placement	Close to Destination	Close to Source
Complexity	Simple	Complex

Extended ACL only is used in firewall Standard ACL is used in routing and switching but not in the firewall

Types of firewall – packet filtering , application and stateful inspection

Packet filtering firewall works from layer 3 to layer 4 it filter based on source ,destination and port number. It is also called as stateless firewall because it doesn't create any entry of the traffic coming to it.

Application firewall works at layer 7 only it filters URL based traffic. Example is proxy

Stateful works from layer 3 to layer 7. It filters the traffic based on source, destination and port. It creates an entry of all the traffic. In stateful firewall, state cocnsists of source , destination IP & port ,ideal timeout and number of bytes packet size .It maintains a state table which is a list of all the active connection.

NGFW is a advanced firewall which offer the deep packet inspection, application awareness and integration intrusion prevention. It protects threat by inspecting the data within the packets ,by managing application traffic and blocking malicious activity.

Key components – deep packet inspection, Application Awareness and control , IPS , threat intelligence and SSL inspection user identity integration

Checkpoint is the first one who gave the stateful inspection firewall

Palo alto was invented in 2005. Named after city San Francisco

In Palo alto we have features like app id, user id and content id.

SP3 architecture- Single-Pass Parallel Processing (SP3) architecture.

Console port is used to gain the physical access of the device.

Management port in the Palo alto is used to manage the Palo alto firewall through https, cli, GUI.

192.168.1.1/24 by default GUI is accesses by the Ip

Firewall works on zone inside (trust zone), outside (untrusted zone), DMZ.

DMZ-

telnet and ssh is to get remote access.
we prefer ssh since it encrypts and sends the password while telnet sends as a plain text.

SNMPv1- Authentication is community strings(plaintext)
Security- minimal vulnerable to attacks.

Operations – get, set, trap

SNMPv2- Authentication is community strings(plaintext)
Security- minimal vulnerable to attack

Operations – get, set, trap, getbulk, inform

SNMPv3- Authentication is user based authentication(USM)
Security- strong security with encryption

Operations – get, set, trap, getbulk, inform

FQDN-

Panorama is used to centralised tool to manage all the palo alto

Since it is cenralised I can do all the configuration in the panaroma

Palo alto deployment mode:

1.tap mode (for monitoring)

2.v-wire mode (transparent just forwards but inspects app,content,user id,encrypted traffic)

3. l3mode (mostly layer 3, routed mode mostly as the deployment mode)

Used for routed traffic (interfaces have IP addresses).

4. l2 mode (mac & Used when the firewall acts like a switch (no IP addresses on interfaces).

Single Pass Parallel Processing (SP3) Architecture

Palo Alto Networks next-generation firewalls are based on a unique Single Pass Parallel Processing (SP3) Architecture – which enables high-throughput, low-latency network security, even while incorporating unprecedented features and technology.

Palo Alto Networks solves the performance problems that plague today's security infrastructure with the SP3 architecture, which combines two complementary components:

- **Single Pass software**
- **Parallel Processing hardware**

The results is the perfect mix of raw throughput, transaction processing and network security that today's high performance networks require.

The second important element of the Parallel Processing hardware is the use of discrete, specialized processing groups that work in harmony to perform several critical functions.

- **Networking: routing, flow lookup, stats counting, NAT, and similar functions are performed on network-specific hardware**

- **User-ID, App-ID, and policy all occur on a multi-core security engine with hardware acceleration for encryption, decryption, and decompression.**
- **Content-ID content analysis uses dedicated, specialized content scanning engine**
- **On the control plane, a dedicated management processor (with dedicated disk and RAM) drives the configuration management, logging, and reporting without touching data processing hardware.**

When packet comes for the first time it is slow path
when comes again like whether session is available then fastpath

Dividing 1 physical firewall into multiple virtual firewalls is called virtual system

App-ID enables you to see the applications on your network and learn how they work, their behavioural characteristics, and their relative risk. Applications and application functions are identified via multiple techniques, including application signatures, decryption (if needed), protocol decoding, and heuristics. This allows granular control, for example, allowing only sanctioned Office 365 accounts, or allowing Slack for instant messaging but blocking file transfer.

Content-ID in Palo Alto Networks Next-Generation Firewalls is a security feature that scans allowed traffic in real-time for threats, malware, vulnerabilities, and data leaks. As a core pillar alongside App-ID and User-ID, it uses a stream-based engine to provide full content inspection across all ports and protocols.

Key Aspects of Content-ID:

- Threat Prevention: Identifies and blocks viruses, spyware, vulnerability exploits, and malicious files.
- Data Loss Prevention (DLP): Monitors for and blocks the transfer of sensitive data using predefined or custom data patterns.
- URL Filtering: Controls access to malicious or unapproved websites.
- Single-Pass Architecture: Analyses traffic once, enabling, analysing, and protecting simultaneously without significantly affecting performance.
- Configuration: Content-ID security profiles (Antivirus, Anti-Spyware, Vulnerability Protection, File Blocking) are applied to security policies.
- Settings Management: Located at Device > Setup > Content-ID in the web interface

User-ID™ enables you to identify all users on your network using a variety of techniques to ensure that you can identify users in all locations using a variety of access methods and operating systems, including Microsoft Windows, Apple iOS, Mac OS, Android, and Linux®/UNIX. Knowing who your users are instead of just their IP addresses enables:

Visibility

Policy control

Logging, reporting, forensics

Acc tab is the graphical representation of the traffic

Application Override policies bypass layer 7 processing and threat inspection and instead use less secure stateful layer 4 inspection. Application Override policies prevent the firewall from performing layer 7 application identification and layer 7 threat inspection and prevention; do not use Application Override unless you must. Instead, create a custom application or create a custom service timeout so that you maintain visibility into, control, and inspect the application in regular layer 7 Security policy rules.

Panaroma looks alike like palo alto and it have extra tab named panaroma where I can see all the connected devices

Panorama ->manage device->summary->add with serial number and auth key

Then palo alto device->panorama setting using same auth key used in panorama

Add new device , then generate authentication key using device registration auth key

Palo Alto Panorama device groups and templates **enable centralized management of firewalls. Device groups handle logical, policy-based configurations (rules, objects), while templates manage network and system configurations (interfaces, routing, zones).**

Device group enables us to add firewall to a group where I can apply rules to group which applies to all the firewall included in the group added using serial number of both h/w and s/w

Both primary and secondary firewall have same ip
template for network configuration.

Palo alto version cannot be higher than the panorama version and it can be equal to or less than palo alto version.

Policy Rules – pre , post and default rule

Pre rule that are pushed from the panaroma to firewall and evaluated before local firewall rule.

Local rules are pushed directly on the firewall and evaluated after the pre rule and post rule.

Post rule that are pushed from the panaroma but evaluated after the local firewall rule.

Default rule there are 2 rules

Inter zone denies traffic between different zone (Trust to untrust is blocked)

Intra zone (same zone) allow traffic between the same zone

Security policy rule – name, source – trust,untrust,edge and ip , destination -zone, address ,

Service/url category – can add service if can't find can add service manually

We can also see who made the commits

I can enable, disable , edit ,modify , delete rules

Admin roles – which type of role I can assign to like enable , read only and disable

Read , read write ,super

Audit admin, security admin, crypto admin

5518 port is used by splunk to communicate with panorama

Protocol no of TCP 6 , UDP 17

Monitor tab is used to monitor network traffic for viewing , analyzing and reporting.

Monitor -> session browser

It is used to see the state table entries i.e traffic coming

Packet capturing available only in palo alto

Packet capture need to be off since it consumes amount of firewall.

Where we can configure the filtering for packet capture filtering

Stage-drop ,firewall,receive and packet count-100

10.241.67.174

10.241.80.72

Syslogs available only in palo

Device backup:

Normal

Snapshot

Running config

Candidate config are done on fw but not pushed to

Routing:

Network->virtual router ->static routes

Static -1 is default metric

Eigrp is not supported by the palo alto firewall

The Routing Information Protocol (RIP) is designed to help routers determine the best path for sending data packets across a network. It uses hop count as its routing metric and is primarily used in small to medium-sized networks due to its scalability limitations. max hop count is 15 and 16 is unreachable.

- Every 30 seconds, routers exchange their complete routing tables with neighbours using periodic updates.

OSPF- Open Shortest Path First (OSPF) is a link-state Interior Gateway Protocol (IGP) developed by the Internet Engineering Task Force (IETF) to provide efficient routing within an Autonomous System (AS). OSPF uses the Shortest Path First (SPF) algorithm, developed by Dijkstra, to determine the best route to each destination.

Open Shortest Path First (OSPF) is a link-state routing protocol used to find the best path between routers within an Autonomous System (AS). OSPF routers establish neighbor relationships to exchange routing information, and these relationships play a crucial role in the formation of OSPF adjacencies. Understanding the transitions between OSPF protocol states is essential for optimizing network performance and troubleshooting.

Note: Unlike distance-vector protocols (like RIP), OSPF does not send periodic updates, instead, it triggers updates only when a change occurs in the network. This makes OSPF faster, more scalable and more efficient, making it widely used in large enterprise and service provider networks.

AD of ospf-110, protocol no-89

224.0.0.5: *All OSPF routers.*
224.0.0.6: *Designated Router (DR) and Backup Designated Router (BDR).*

Areas

- **Backbone Area (Area 0)**: Core of the OSPF domain; all other areas must connect here.
- **Normal Areas**: Standard areas that connect to Area 0.
- **Stub Areas**: Reduce routing table size by limiting external route information.

Router Types

- **Internal Router:** Operates inside a single area.
- **Area Border Router (ABR):** Connects multiple areas to Area 0.

- **Autonomous System Boundary Router (ASBR):** Connects OSPF to other routing protocols.
- **Designated Router (DR):** Elected on broadcast/multi-access networks to reduce adjacencies.
- **Backup Designated Router (BDR):** Backup for DR to ensure redundancy.

OSPF Message Types

1. **Hello Message**: Discovers and maintains neighbor relationships.
2. **Database Description (DBD)**: Summarizes LSDB contents for comparison.
3. **Link-State Request (LSR)**: Requests missing LSAs from neighbors.
4. **Link-State Update (LSU)**: Contains full LSAs to update neighbors.
5. **Link-State Acknowledgment (LSAck)**: Confirms receipt of LSUs.

OSPF Timers

- **Hello Timer**: Interval between Hello packets (default: 10 sec).
- **Dead Timer**: Time to declare a neighbor down if Hellos are missing (default: 40 sec).
- **Adjustable Timers**: Can be modified for faster or slower detection depending on the network.

Falg- a:active,s-static,c-connected, h-host

The 7 States

1. Down State

- No hello packets received
- Neighbor is **not known yet**

---

2. Attempt State (NBMA only)

- Router is actively sending hello packets
- Waiting for reply
- Used in **Non-Broadcast networks (NBMA)**

3. Init State

- Router received hello packet
- But **its own ID is NOT in neighbor list**

---

4. 2-Way State

- Both routers see each other
  ✅ Bidirectional communication established

📌 DR/BDR election happens here (in broadcast networks)

---

5. ExStart State

- Routers decide:
  - Who is **Master**
  - Who is **Slave**
- Begin exchange process

---

6. Exchange State

- Routers exchange **DBD (Database Description) packets**
- Share summarized LSDB info

---

7. Loading State

- Routers request missing LSAs
- Using:
  - LSR (Link State Request)
  - LSU (Link State Update)

---

8. Full State ✅

- LSDBs are fully synchronized
- Adjacency completed

For cli use putty then put host address- show system info command, show routing route to get routing info and session flow can be seen using command show session all.

broadcast is a network addressing where a single ip is shared across multiple geo this server

Objects : 2 group address

Object group is an organisational tool within the objects tab used to bundle individual objects , simplifying security policy creation, management and readability.

Session Browser

The Session Browser is a real-time view of the firewall's current "active" memory. While the Traffic Logs show you what *already happened* (history), the Session Browser shows you what is *happening right now*.

What it tells you:

- Source/Destination: IP addresses and ports.
- Application: What App-ID the firewall has identified (e.g., web-browsing, ssl, or facebook).
- State: Is the session "Active," "Discard" (blocked), or "Closing"?
- Flags: Specifically the "A" flag (Active) or the "D" flag (Discard).
- NAT: You can see if a session is currently being translated via U-Turn NAT or standard NAT.

Where to find it: Go to Monitor > Session Browser.

Pro-Tip: If a user says "I can't get to the website right now," check the Session Browser. If you see the session with a state of Discard, you know the firewall is actively dropping it.

Packet Capture (PCAP)

Packet capturing is the process of recording the "raw" data packets for deep analysis.

Does it use Wireshark?

The answer is both No and Yes.

- No: The firewall itself does not "run" the Wireshark application. It records the traffic in a standard format called .pcap.
- Yes: Because the firewall saves the files as .pcap, you download those files from the firewall and open them in Wireshark on your laptop to see the "gory details" (hex code, every single header, and data payload).

Day-4

NAT-Static, dynamic, PAT (Port address translation)

Static NAT is one to one mapping and bidirectional in nature where a private, internal IP address is permanently mapped to a specific public, external IP address.

Dynamic NAT is like many to few that means we are creating a NAT pool where source IP is taking the public IP and communicating outside. Once pool is exhausted natting will not be possible. Dynamic NAT is bidirectional in nature.

PAT (Port address translation) -is a networking technology that allows multiple devices on a local area network (LAN) to share a single public IP address to access the internet. It works by translating private IP addresses to a public one, appending a unique source port number to each session to differentiate traffic and manage return data, effectively conserving IPv4 addresses

In palo alto there are 3 types of NAT – source , destination, u-turn

Source NAT- Dynamic ip and port(working same as PAT), Dynamic IP, Static IP

1. Dynamic IP and Port (DIPP)

This is the most common form of NAT, often referred to as **PAT (Port Address Translation)**.

- **How it works:** It maps multiple internal private IP addresses to a single public IP address (or a small pool of addresses). To distinguish between different internal users, the firewall assigns a unique **Source Port** to each session.
- **Capacity:** A single IP can support approximately 64,000 concurrent sessions.
- **Use Case:** Standard internet access for an entire office using only one or two public IP addresses.

---

2. Dynamic IP

This is a "pool-based" NAT without port translation.

- **How it works:** It maps an internal private IP to the next available public IP in a defined pool. Once an internal IP is assigned a public IP from the pool, that public IP is reserved for that user's sessions.
- **Constraint:** Unlike DIPP, it does **not** use port translation. This means if you have 10 public IPs in your pool, only 10 internal users can go online at the exact same time. If an 11th user tries to connect, the request is dropped until a pool address becomes free.
- **Use Case:** Rare. Used for specific legacy applications that are "NAT-sensitive" and fail if the source port is modified.

---

3. Static IP

This is a permanent, one-to-one fixed mapping.

- **How it works:** A specific internal private IP is manually and permanently mapped to a specific public IP. No other device can use that public IP, and the mapping never changes.
- **Directionality:** While this is a "Source NAT" rule, a Static IP NAT in Palo Alto is often configured as **Bi-directional**, meaning it automatically

creates the corresponding Destination NAT to allow traffic back into that server.

- **Use Case:** High-priority servers (like a Mail Server or Web Server) that need a consistent identity for external communication.

Summary Table

NAT Type	**Mapping Ratio**	**Port Translation?**	**Primary Benefit**
DIPP (PAT)	Many-to-One	**Yes**	Maximum IP efficiency.
Dynamic IP	One-to-One (Pool)	No	Maintains original source ports.
Static IP	One-to-One (Fixed)	No	Permanent identity for a specific server.

Destination NAT- performed on incoming packets

How it works

The firewall intercepts a packet, looks at the **Original Destination IP**, and swaps it for a **Translated Destination IP**. This happens *before* the security policy check, but the firewall still uses the "Pre-NAT" destination zone to determine which security rule applies.

Types of Destination NAT

1. Static Destination NAT (1-to-1)

- **What it is:** A permanent, one-to-one mapping.

- **How it works:** Any traffic hitting a specific Public IP is automatically sent to a specific Internal IP.
- **Use Case:** A dedicated public IP address assigned to a single internal server.

2. Port Forwarding (Port Translation)

- **What it is:** Mapping a specific port on a public IP to an internal IP.
- **How it works:** You translate both the **IP** and the **Port**.
- **Example:** Traffic hitting your Public IP on port **8080** is translated to your Internal Server on port **80**.

3. Dynamic Destination NAT (One-to-Many / Load Balancing)

- **What it is:** Mapping one public IP to a pool of internal servers.
- **How it works:** The firewall uses an algorithm (like Round Robin) to distribute incoming requests across multiple internal hosts.
- **Use Case:** Basic load balancing for a small farm of web servers.

Summary Table

Type	**Translates**	**Purpose**
Static DNAT	IP only	Dedicated 1:1 server access.
Port Forwarding	IP + Port	Accessing specific services (e.g., RDP, HTTP).
Dynamic DNAT	IP (Pool)	Simple load distribution acr

Creating NAT steps:

Policies->NAT->in general give name then original packet – source zone, destination zone, source address then translated packet – source address translation, destination address translation

Prerequisites to build the HA:

To build HA both devices must be having same model number, same version, same set of licenses and both firewall must have same set of interfaces.

HA stands for **High Availability**. Its goal is simple: to ensure that a system or service remains operational and accessible even if a component (like a cable, a switch, or an entire firewall) fails.

In the context of **Palo Alto Networks**, HA refers to connecting two firewalls together so that if one "dies," the other takes over immediately without dropping your internet connection.

Palo alto HA deployment modes:

1.Active passive

2.active active

Active/Passive (A/P):

- **How it works:** One firewall is actively processing traffic while the other is in a standby (passive) state. The standby unit continuously synchronizes with the active unit.
- **Failover:** If the active unit fails, the passive unit instantly promotes itself to active, taking over the traffic flow using the same policies and configuration.
- **Use Case:** Ideal for organizations prioritizing simplicity and reliable redundancy without needing to manage two active traffic paths.

Active/Active (A/A):

- **How it works:** Both firewalls are active and processing traffic simultaneously. They work synchronously to handle session setup and ownership.
- **Failover:** Both devices continue to process traffic. If one unit fails, the remaining unit absorbs the load.

- **Complexity:** This mode requires advanced network design (e.g., handling asymmetric routing) and is generally more complex to troubleshoot than A/P.
- **Use Case:** Recommended only if you have specific requirements for redundant throughput or handling asymmetric traffic flows.

HA links:

1. HA1(control link) – port 28260 & 28769 , encrypted port – 28 (SSH)

It exchanges hellos, heartbeats, HA state information, and management plane synchronization (routing, User-ID, and configuration updates).

It requires IP.

2. HA2(Data link) - UDP 29281 or IP protocol 99 for HA2 data synchronization

synchronizes session tables, forwarding tables, ARP table, IPSec security associations, and ARP tables between HA peers or from active to passive peers.

3. HA3 link (packet forwarding link):

In addition to ha1 and ha2, an active/active deployment also require a dedicated HA3 link. The firewalls use this link for forwarding packets to the peer during session setup and asymmetric traffic flow.

HA require 4 IP

1.Symmetric Traffic Flow

Symmetric flow occurs when the request (client to server) and the response (server to client) pass through the same firewall and the same interfaces.

2. Asymmetric Traffic Flow

Asymmetric flow occurs when the request takes one path, but the response takes a different path that bypasses the original firewall.

1. Failover

Failover is the process where the passive (or secondary) firewall detects a failure on the active (or primary) firewall and assumes the responsibility of securing traffic.

- **Why it happens:** It is triggered by hardware failure, a loss of the control link heartbeat, administrative suspension, or failure of a monitored object (via link or path monitoring)

2. Link Monitoring

Link Monitoring allows the firewall to watch the physical health of its interfaces.

- **How it works:** You define a "Link Group" consisting of one or more physical interfaces. You specify a failure condition (e.g., "Any" interface in the group goes down, or "All" must go down).
- **Impact:** If the monitored interfaces go down, the firewall registers a link failure. Depending on your configuration, this can trigger a failover to the healthy peer.

3. Path Monitoring

Path Monitoring is a higher-level health check that monitors the "path" through the network to a mission-critical destination (like an upstream router or a core server). 10 ICMP fail

- **How it works:** The firewall sends ICMP pings to a specific destination IP. If those pings fail consistently (after a set number of attempts), the firewall considers the path to be "unreachable."
- **Impact:** Even if the firewall's physical interfaces are "up," the firewall might be isolated from the rest of the network. Path monitoring detects this and triggers a failover to a firewall that still has connectivity.

4. Priority and Pre-emption

These settings determine *which* firewall should be the "Active" unit when both are healthy.

- **Priority:** Each firewall is assigned a numerical priority value. The firewall with the **lower numerical value** (e.g., 50 vs. 100) is preferred to be the Active unit.
- **Preemption:** This is a setting that forces a "failback."
    - If preemption is **disabled** (default), the firewall that took over after a failure remains Active even after the original primary unit recovers.
    - If preemption is **enabled**, the original primary unit will automatically reclaim the "Active" role as soon as it recovers and becomes healthy.

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5. Backup in HA

In the context of HA, "backup" usually refers to two different concepts:

- **Configuration Backup:** It is critical to manually back up the configuration (XML file) of **both** firewalls in an HA pair independently. Even though they synchronize, having a separate export of each firewall's local state is essential for disaster recovery.
- **HA Backup Links:** You can (and should) configure redundant HA links. For instance, if you have a primary HA1 (control) link, you can add a backup HA1 link. If the primary link cable is unplugged or damaged, the firewalls will use the backup link to maintain their heartbeat, preventing an accidental "split-brain" failure where both firewalls try to act as "Active" simultaneously.

Day-5

Scenario: a person is unable to access a application.

First get remote access , ask questions like were u able to access before or when was he accessed last time.

If TCP reset from the sever

TCP reset from the source

Erased out i.e server low

A user trying to application and the traffic is going from firewall but cant come ans- maybe routing issues

Security profiles:

1. **Antivirus**-detects and blocks the virus, worms and trojans.it scans protocols like http, ftp, smtp, IMAP, pop3.Integrate with wildfire for advanced detection.

2. **Anti Spyware**- Prevent the infected system from communicating with command and control servers. It includes DNS sinkhole to stop/protect malicious domain

3. **Vulnerability protection-** It blocks the exploits attempts , protect against the buffer overflow and code execution attacks.

4. **URL filtering-** control website access based on categories. Social media malware, adult, ransomware etc. It can block risky.If URL is blocked and by app rule the app is allowed since its URL is blocked. We can't access the app.

5. **File blocking –** blocks and controls file transfer. Actions- block, continue and alert.

6. **Wildfire analysis**- sends the unknown file to wildfire cloud. Detects the zero-day malware and updates the signature

7. **DLP-** prevents sensitive data leakage. By using confidential keywords and so on.

8. **Data filtering-** protects the zone. Detects port scan

9. **Mobile network protection-** any data coming from mobile network is protected

Feature	**Primary Purpose**
1. Antivirus	Scans files for known malicious signatures to detect and block viruses, worms, and trojans.
2. Spyware Protection	Detects and blocks communication from spyware (software that steals personal information or monitors user activity).
3. Vulnerability Protection	Blocks attempts to exploit known software or hardware vulnerabilities before a patch can be applied.
4. URL Filtering	Controls access to websites based on categories or reputation, preventing users from visiting malicious or inappropriate sites.

Feature	Primary Purpose
6. File Blocking	Restricts specific file types (e.g., executables, scripts, or password-protected archives) from being uploaded or downloaded.
7. WildFire Analysis	A cloud-based sandbox service that executes suspicious, unknown files in a secure environment to observe their behavior before allowing them on the network.
8. DLP (Data Loss Prevention)	Monitors and prevents unauthorized data exfiltration by identifying sensitive patterns (like credit card numbers or PII) in traffic.
9. DOS (Denial of Service)	Protects network resources from being overwhelmed by flood-based attacks designed to crash services or consume bandwidth.
10. Mobile Network Protection	Extends security policies to mobile devices, ensuring traffic from roaming or remote mobile devices is inspected and filtered.

Policy based routing in the Palo alto firewall let you override the normal routing table and force specific traffic to take different path based on defined condition (source IP, destination IP, application, user and service)

PBR=match traffic ->override routing->send via chosen hop

MPLS-multi protocol layer system

VPN:

It is used to communicate over insecure medium in a secure manner.

It works on confidentiality, integrity and auntenticity.

Classification of VPN:

Intranet VPN- same company communicating over leased lines / intranet lines

Extranet VPN- different companies communicating over internet.

Depending on Deployment:

Site to site-

S2S protocols allow an organization to establish secure communication between two or more sites/office so that it can send traffic using shared medium such as InternetGRE, IPSec

Remote access- remote access protocol benefits an organisation SSL

Depending on Internet Engineering Task Force (IETF) standards:

L2TP, L2F, PPTP, MPLS, GRE, IPsec, SSL

Aim of any VPN:

Confidentiality (Encryption – Data protection)

Symmetric encryption like DES, 3DES, AES.

Integrity (hashing- detect data modification)

Authentication(PKI), non-repudiation, anti replay

DAY-6

WAF-Web Application Firewall

It works on the layer 7 which is cloud or on premises

It protects web app by filtering and monitoring the HTTP traffic.

Acts as the barrier between client and web server, inspecting all the incoming traffic to prevent attacks before they reach web app.

Key purposes of WAF:

1. Protection against OWASP top 10
2. Traffic filtering
3. Application layer defence
4. Rate limiting and traffic management
5. Zero-day protection
6. Compliance and security standards

1. Cloud-Based WAF (SaaS/Managed)

In this model, your web traffic is redirected through the provider's global network (via DNS change or BGP routing) before it ever reaches your server.

- **Deployment:** Very simple. You typically update your domain's DNS (A or CNAME record) to point to the provider's network.
- **Advantages:**
    - **Scalability:** Automatically handles massive traffic spikes without local hardware upgrades.
    - **Zero Maintenance:** The provider manages updates, threat intelligence, and patching.
    - **Edge Protection:** Blocks attacks at the "edge" (closer to the attacker), preventing your origin server from being overwhelmed.
- **Disadvantages:**

    - **Latency:** Adding an extra hop to a third-party data center can add a few milliseconds to connection times.
    - **Data Privacy:** Traffic is decrypted and inspected by a third party, which can be a compliance issue for some regulated industries.

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2. On-Premises WAF (Hardware/Virtual)

You install a physical appliance or a virtual machine (VM) inside your own data center, directly in front of your web servers.

- **Deployment:** You manually route traffic from your load balancer or edge router into the WAF appliance. You are responsible for installation, cabling, and network configuration.
- **Advantages:**
    - **Data Sovereignty:** Traffic never leaves your premises. Full control over security logs and data.
    - **Performance:** Lowest possible latency for internal or locally hosted traffic.
    - **Customization:** Full administrative access to fine-tune rules specific to your proprietary application.
- **Disadvantages:**
    - **Maintenance Burden:** Your team must handle firmware updates, capacity planning, and hardware failure.
    - **High Upfront Cost:** Requires capital expenditure for hardware or software licensing.

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3. Hybrid / Cloud-Native WAF

This deployment is popular for modern cloud-hosted applications (like AWS WAF, Azure WAF, or Google Cloud Armor).

- **Deployment:** The WAF is integrated directly into your cloud provider's load balancer or Application Delivery Controller (ADC).
- **Advantages:**

  - **Seamless Integration:** Native "tick-box" activation within the cloud console.
  - **Elasticity:** Scales perfectly with your cloud instance (e.g., if your web servers auto-scale, the WAF scales with them).
- **Disadvantages:**
  - **Vendor Lock-in:** Moving your app to another cloud provider might mean re-configuring your WAF policies from scratch.

Summary Comparison Table

Deployment	Latency	Maintenance	Security Control
Cloud-Based	Moderate	Very Low	Limited (Provider manages)
On-Premises	Lowest	High	Maximum
Cloud-Native	Low	Low	High

Barracuda:

The Barracuda Web Application Firewall (WAF) is a comprehensive security solution designed to protect web applications, APIs, and mobile app backends from sophisticated cyberattacks. It acts as a gatekeeper, sitting between the internet and your web servers to filter out malicious traffic before it reaches your data

It is a robust solution that secure web applications, api, and web server from cyber-attacks.

Key capabilities:

1. Advanced threat protection
2. Automated security updates
3. Bot protection

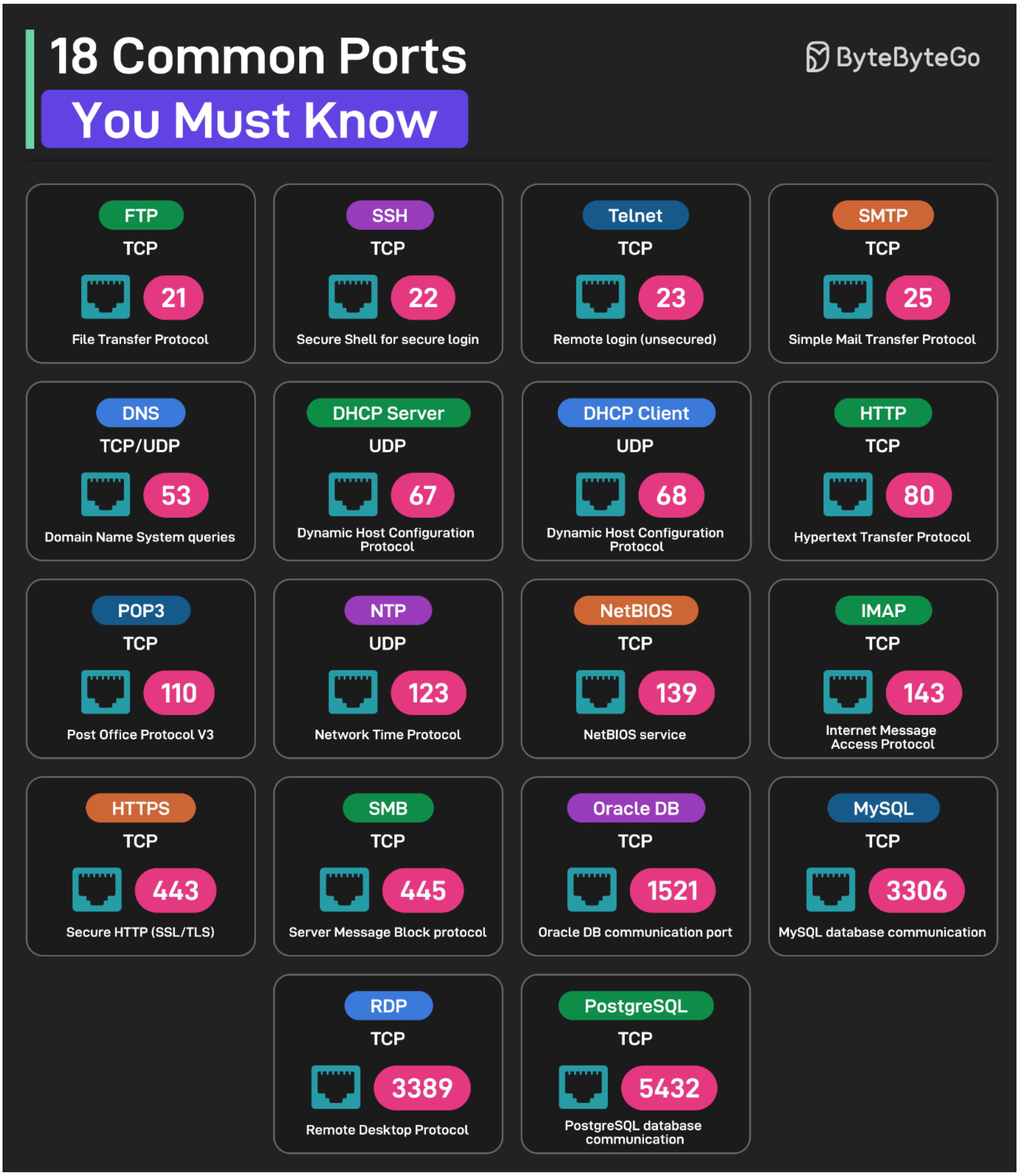
4. SSL/TLS offloading
5. API security
6. Web application acceleration

Deployment:

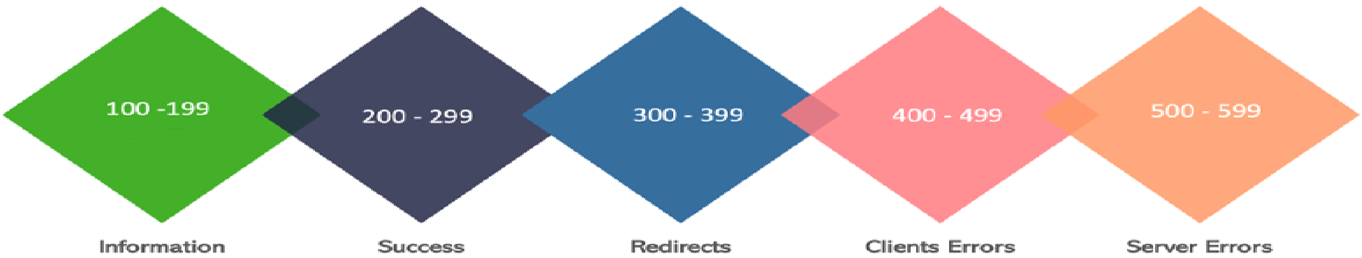
1. Cloud
2. On premises
3. Hybrid

161,162- SNMP

20,21 – FTP (data, control)



18 Common Ports
You Must Know
ByteByteGo
FTP
TCP
21
File Transfer Protocol
SSH
TCP
22
Secure Shell for secure login
Telnet
TCP
23
Remote login (unsecured)
SMTP
TCP
25
Simple Mail Transfer Protocol
DNS
TCP/UDP
53
Domain Name System queries
DHCP Server
UDP
67
Dynamic Host Configuration Protocol
DHCP Client
UDP
68
Dynamic Host Configuration Protocol
HTTP
TCP
80
Hypertext Transfer Protocol
POP3
TCP
110
Post Office Protocol V3
NTP
UDP
123
Network Time Protocol
NetBIOS
TCP
139
NetBIOS service
IMAP
TCP
143
Internet Message Access Protocol
HTTPS
TCP
443
Secure HTTP (SSL/TLS)
SMB
TCP
445
Server Message Block protocol
Oracle DB
TCP
1521
Oracle DB communication port
MySQL
TCP
3306
MySQL database communication
RDP
TCP
3389
Remote Desktop Protocol
PostgreSQL
TCP
5432
PostgreSQL database communication



100 -199
200 - 299
300 - 399
400 - 499
500 - 599
Information
Success
Redirects
Clients Errors
Server Errors

Adding a Palo Alto Networks firewall to Panorama involves a handshake between the device and the management platform. Think of it as a "mutual introduction" where both sides need to know who the other is before they start talking.

Here is the streamlined process to get it done.

Phase 1: Prepare the Firewall

Before Panorama can take control, the firewall needs to know where to look.

1. **Login** to the Firewall web interface.
2. Navigate to **Device > Setup > Management**.
3. Edit the **Panorama Settings** widget.
4. Enter the **IP addresses** of your Panorama server (Primary and Secondary, if applicable).
5. **Commit** the changes to the firewall locally.

Phase 2: Add the Device to Panorama

Now, you need to tell Panorama to expect a connection from this specific serial number.

1. **Login** to the Panorama web interface.
2. Navigate to **Panorama > Managed Devices > Summary**.
3. Click **Add** at the bottom of the page.
4. Enter the **Serial Number** of the firewall (this is the unique identifier Panorama uses).
5. (Optional) Uncheck "Import classic objects" if you want a clean slate, or keep it if you plan to pull existing rules into Panorama later.
6. Click **OK**.

Phase 3: Establish Connectivity (Auth Key)

For security, modern PAN-OS versions require an **Authentication Key** to prevent unauthorized devices from joining.

1. On Panorama, go to **Panorama > Managed Devices > Summary**.

2. Click **Registration Auth Key** at the bottom.
3. Generate a new key and **copy it**.
4. Go back to the **Firewall** interface under **Device > Setup > Management > Panorama Settings**.
5. Paste the key into the **Registration Auth Key** field.
6. **Commit** on the Firewall.

Phase 4: Finalize on Panorama

The final step is to push the configuration from Panorama to acknowledge the new device.

1. On Panorama, click **Commit** in the top right.
2. Select **Commit to Panorama**.
3. Once the commit finishes, check the **Connected** column under **Managed Devices**. It should turn **Green**.

Quick Troubleshooting Tips

- **Port 3978:** Ensure that TCP port 3978 is open between the Firewall's Management IP and Panorama. This is the "heartbeat" port.
- **Time Sync:** Ensure both Panorama and the Firewall have the correct time (use NTP). If the certificates are off by even a few minutes, the SSL handshake will fail.
- **Serial Number:** Double-check for typos; even one wrong digit will keep the status "Disconnected."

Are you planning to import existing rules from this firewall into a new Template Stack, or are you starting with a fresh configuration?

CLI commands

To ace a Palo Alto Networks interview, you need to demonstrate that you can navigate the CLI beyond just basic configuration. Interviewers love to see that you know how to **troubleshoot** and **verify** traffic flow.

The CLI has two modes: **Operational** (denoted by >) for viewing status, and **Configuration** (denoted by #) for making changes.

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1. Traffic & Session Troubleshooting

These are the "bread and butter" commands for identifying why a specific connection is failing.

- show session all filter source <IP>: Shows all active sessions for a specific source.
- show session id <ID>: Provides granular details about a specific session, including NAT translations and security rules applied.
- test security-policy-match source <IP> destination <IP> protocol <#> destination-port <#>: **Crucial.** This simulates a packet to see exactly which security rule it hits without actually sending traffic.
- test routing-table-lookup ip <IP>: Verifies which route (and egress interface) the firewall will use for a specific destination.

2. Interface & System Health

Use these to verify the physical and logical health of the box.

- show interface all: Gives a quick table view of all interfaces, their status (up/down), and zones.
- show interface hardware <name>: Checks for physical errors, collisions, or duplex mismatches.
- show system resources: The Palo Alto version of "top." It shows CPU and memory utilization for DP (Data Plane) and MP (Management Plane).
- show system info: Displays the serial number, software version (PAN-OS), and uptime.

3. Management & Processes

If the GUI is slow or a specific feature (like User-ID) isn't working, these commands are your best friends.

- debug software restart process management-server: Restarts the web interface and management logic without dropping traffic.
- show jobs all: Shows the status of commits, upgrades, or content installs.
- tail follow yes mp-log ms.log: Real-time viewing of the management server log (great for troubleshooting "Commit Failed" errors).

4. High Availability (HA)

If you're interviewing for a production environment role, HA is guaranteed to come up.

- show high-availability state: Shows if the device is Active, Passive, or Non-functional.
- show high-availability all: Provides a deep dive into synchronization status and heartbeats.
- request high-availability state suspend: Manually forces a failover by putting the current active unit into a suspended state.

5. GlobalProtect & User-ID

For remote access and identity-based policy questions.

- show globalprotect-gateway current-user: Lists all users currently connected via VPN.
- show user ip-user-mapping all: Shows the table where the firewall maps an IP address to a specific username.

Pro-Tip for the Interview

When an interviewer asks "How do you troubleshoot a blocked connection?", don't just say "check the logs." Start with:

"First, I'd verify the routing and policy match using test security-policy-match. Then, I'd check show session all filter to see if the session is being discarded or if it's stuck in a SYN-sent state."

Which of these areas (VPN, Routing, or Security Policies) do you feel most "shaky" on for your upcoming interview?

commit: Applies all pending changes to the running configuration.

validate: Checks the syntax and consistency of the candidate configuration without applying it (great for catching errors before a commit).

view-config diff: Shows the difference between the running config and your current changes.

request content upgrade check: Checks for new threat and application signatures.

request content upgrade download latest: Pulls the newest signatures from the updates server.

show arp all: Shows the ARP table to verify Layer 2 connectivity.

show routing neighbor dhcp: Useful for checking DHCP-assigned addresses on interfaces.

Show system resources

Show system info

The management plane handles configuration, monitoring, and user access (e.g., CLI, GUI, API). The control plane makes decisions about routing, workload placement, and system state

On a Palo Alto Networks firewall, the architecture is physically split into two distinct parts: the **Management Plane** and the **Data Plane**.

The **Control Plane** is technically a function that lives inside the Management Plane, but in common network discussion, these terms are often used to distinguish how the firewall "thinks" versus how it "manages."

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1. The Management Plane (The "Admin" Brain)

This is the part of the firewall you interact with. It has its own dedicated processor, RAM, and storage (SSD).

- **Function:** Handles all administrative tasks that are not related to high-speed packet processing.
- **Tasks:** * Running the WebUI, CLI, and XML API.
    - Generating logs and reports.
    - Handling configuration commits.
    - Managing device updates (App-ID, Content, Antivirus).
- **Physical Port:** The **MGT** port connects directly to this plane.

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2. The Control Plane (The "Strategy" Brain)

In Palo Alto terminology, the Control Plane is a subset of the Management Plane. It handles the "logic" of networking.

- **Function:** It builds the "map" that the Data Plane uses to move traffic.
- **Tasks:**
    - Running dynamic routing protocols (**OSPF, BGP, RIP**).
    - Calculating the ARP table and MAC table.
    - Handling SSL certificate exchange and handshakes.
    - Communicating with User-ID agents to map IPs to Usernames.

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3. Why the Split Matters (The "Yellow Brick" Concept)

Palo Alto uses a **Single-Pass Parallel Processing (SP3)** architecture. The Management/Control plane is physically separated from the Data Plane (where traffic flows).

- **Stability:** If a massive DDoS attack hits your firewall or a routing loop occurs, your **Data Plane** might be 100% busy, but you can still log into the **Management Plane** via the MGT port to fix the issue.
- **Performance:** The Management Plane doesn't slow down the Data Plane. High-speed traffic doesn't have to wait for the firewall to finish "writing a log" before it can pass through.

Summary Comparison

Feature	**Management Plane**	**Control Plane**
Primary Goal	Administration & Storage.	Routing & Logic.
Interaction	Admin via WebUI/SSH.	Switch-to-Switch/Protocol talk.
Examples	Committing a config, viewing logs.	Learning a BGP route, ARP lookup.
Hardware	Dedicated CPU/RAM.	Shared with Management Plane.

Key Takeaway: The **Control Plane** decides *where* a packet should go (routing), while the **Management Plane** allows *you* to tell the firewall how to behave. Both are separated from the **Data Plane**, which is the "muscle" that actually moves the packets.

Command to do manual failover

request high-availability state suspend